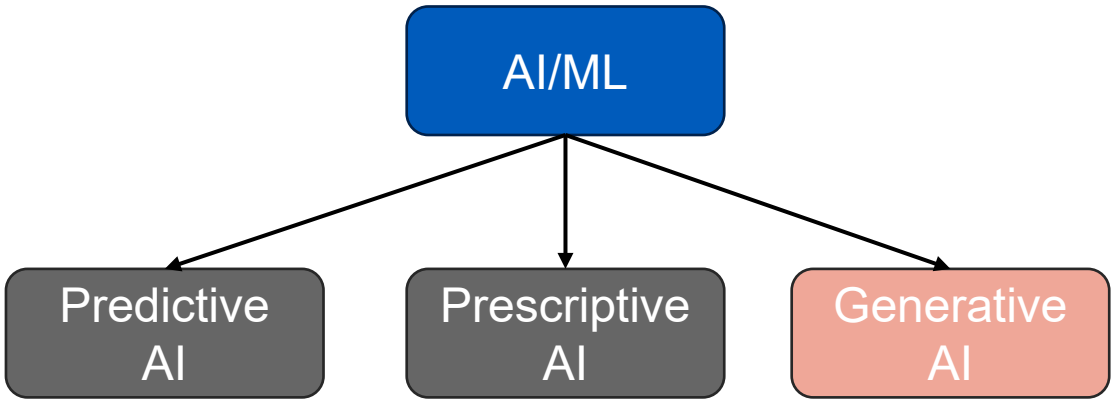
CE 400/500

Special Topics: AI for Chemical Engineers

Lecture 18: Introduction to Generative AI, Hands-on Practice of Variational Autoencoder for Drug Discovery

University at Buffalo
The State University of New York

Generative AI



- Generative AI is changing the way scientists and engineers design, simulate, and communicate. It learns the underlying data distribution and creates new, realistic samples rather than just analyzing existing ones.

- It is widely used in molecule and material generation, catalyst and polymer design, automated synthesis planning, generation of optimal or novel process flow diagrams for chemical plants, and agentic AI for orchestrating autonomous laboratory.

Historical Milestones of Generative AI

1960s: ELIZA, one of the first chatbots, emerged as a primitive generative AI using simple pattern matching. Neural network ideas (perceptron) and expert systems also began here.

1980s-1990s: Recurrent Neural Networks and later Long Short-Term Memory (LSTM) enabled AI to model and generate sequences like text more coherently.

2010s: Advances in GPU and big data fueled deep learning, enabling realistic image, audio, and text generation. Key models: variational autoencoder (2013), generative adversarial networks (2014), diffusion (2015).

2017-2018: The transformer architecture (2017) and GPT models (2018) revolutionized language generation, making large-scale, coherent text generation fast and efficient.

2020s: DALL-E (2021) and GPT-4 (2023) brought text-to-image and advanced language generation into the mainstream, enabling creative applications worldwide.

ELIZA chatbot
(Credit: masswerk.at)

Deepfakes
(Credit: The New York Times)

https://toloka.ai/blog/history-of-generative-ai/

Outline

- Embeddings & latent space
- Major generative ML algorithms: generative adversarial network, diffusion, variational autoencoder, with their example applications in chemical & material engineering
- Hands-on practice of implementing variational autoencoder for drug discovery

Revisit Feature Representation with Numerical Data

All ML models, including generative models, operate on numbers, not words or images directly. The first step is to encode information into vectors (features) that capture meaning and structure.

For example, we have built features with numerical data so far:

Wine quality predictions

Feature	Example Value
fixed acidity	7.4
volatile acidity	0.700
citric acid	0.00
residual sugar	1.9
chlorides	0.076

MOF synthesis conditions

Synthesis variable	Optimisation constraints
Water (H_2O)	0 – 6 ml
Dimethylformamide (DMF)	0 – 6 ml
Ethanol (EtOH)	0 – 6 ml
Methanol (MeOH)	0 – 6 ml
Isopropyl alcohol (iPrOH)	0 – 6 ml
Reactants ratio	0.8 – 1.8
Temperature	100 – 200 °C
Microwave power	150 – 250 W
Reaction time	2 – 60 mins

However, how can we encode **categorical data** into vectors? Such as words like “France”, “Paris” or in car price prediction where feature vector encodes car’s exterior, interior, and seatbelt color, e.g., [red, blue, blue]?

One-Hot Encoding

One-hot encoding is the most basic way to numerically represent categories. Each token or word is represented by a vector of 0s and a single 1.

Word: “cab”

Vocabulary	c	a	b
a	0	1	0
b	0	0	1
c	1	0	0
⋮	0	0	0
z	0	0	0

Flatten to a 1D vector

Example 1: Each character as a token

Features

Vocabulary	red	blue	blue
red	1	0	0
blue	0	1	1
yellow	0	0	0
⋮	0	0	0
orange	0	0	0

Example 2: Each word/color as a token

- Pros: unambiguous;
- Cons: high dimensional, no relationships

Embedding/Latent Space

- An embedding is a vector representation of data in embedding (latent) space.
- Generally speaking, a model finds potential embeddings by projecting the high-dimensional space of initial data vectors (e.g., one-hot encoding) into a lower-dimensional space.
- If well constructed, relative positions of data points in embedding space may have a potential semantic relationship.
- Generative models use this space to interpolate between known examples and generate new ones.

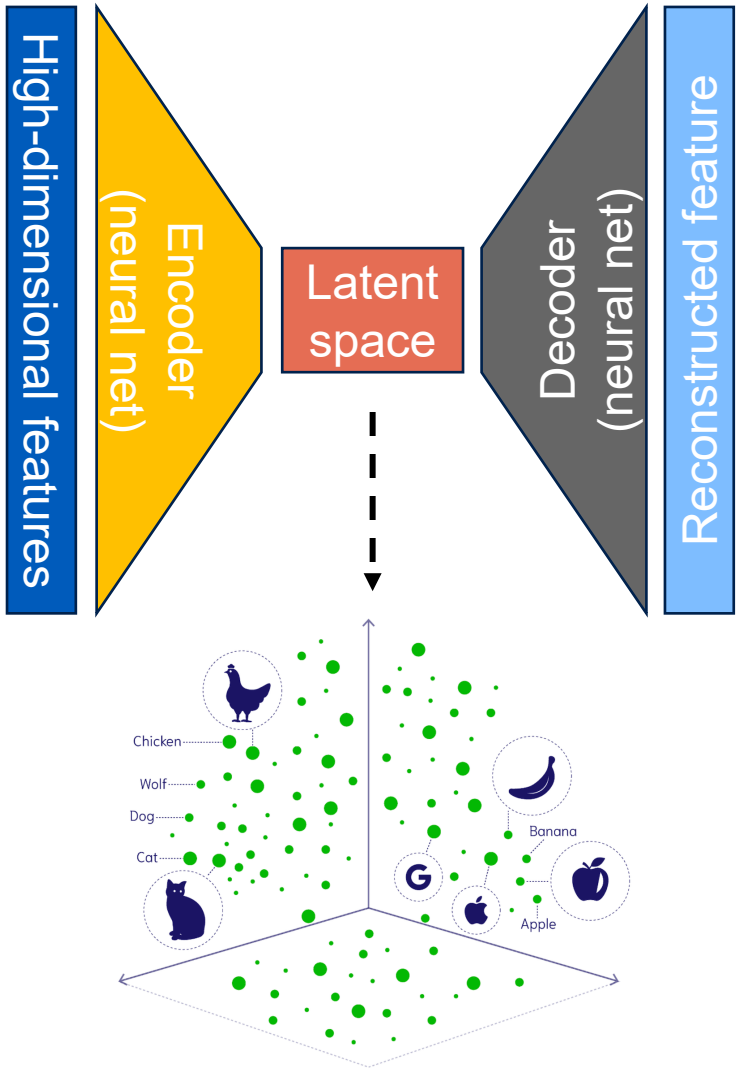
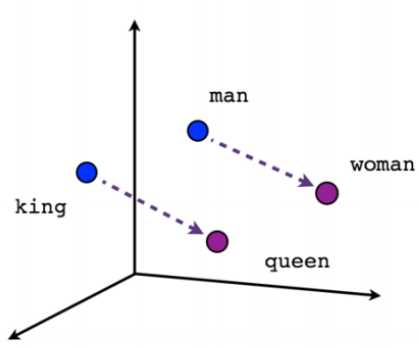
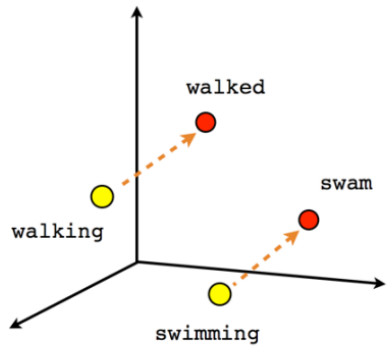


Image credit: weaviate.io

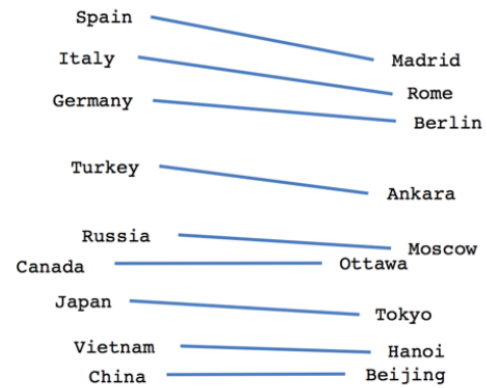
Example: Word Embeddings



Male-Female



Verb tense



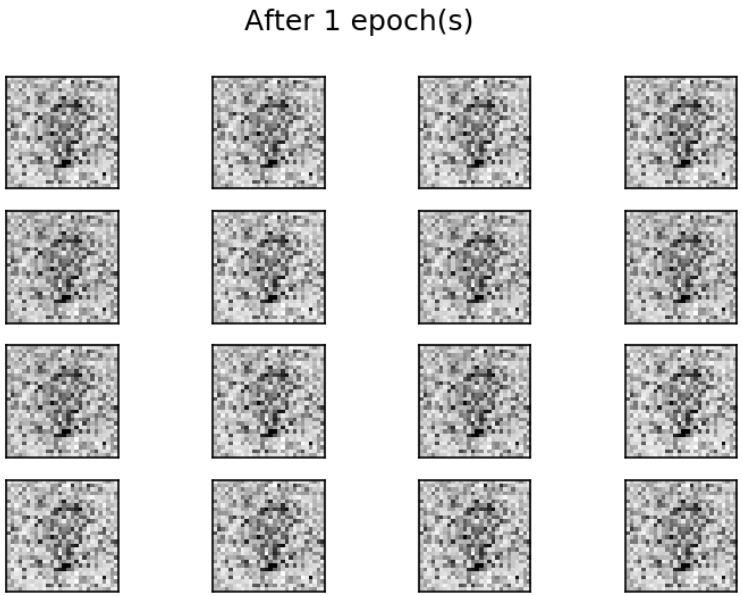
Country-Capital

- While it's an older example, and largely superseded by other models, the **word2vec** model remains useful for illustration. word2vec trains on a corpus of documents to obtain a single global embedding per word. When each word or data point has a single embedding vector, this is called a **static embedding**. Words with similar context appear close together (i.e., smaller distance) in vector space.
- This idea revolutionized language processing and later inspired generative architectures.
- Check out the word2vec embedding space by yourself here!

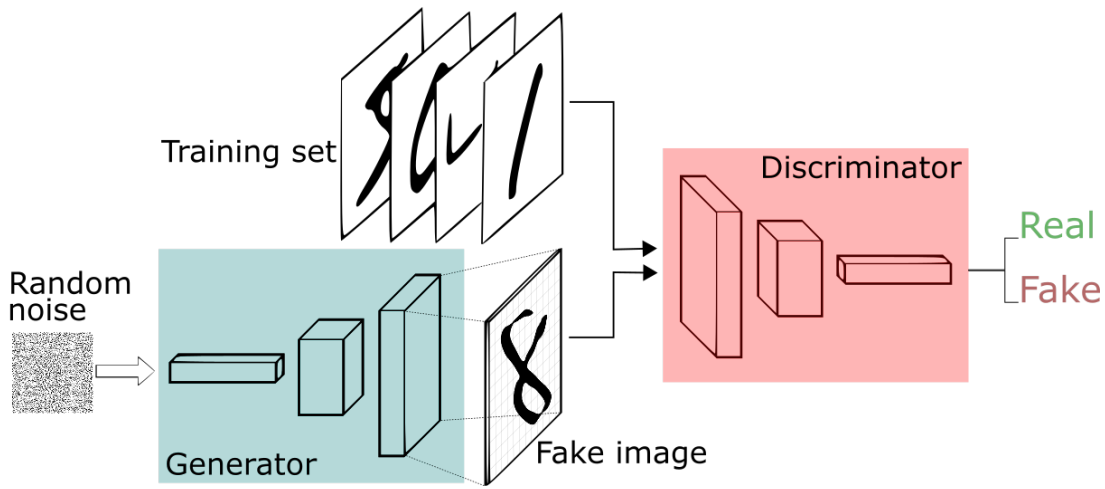
Image credit: Medium

Major Generative ML Algorithms

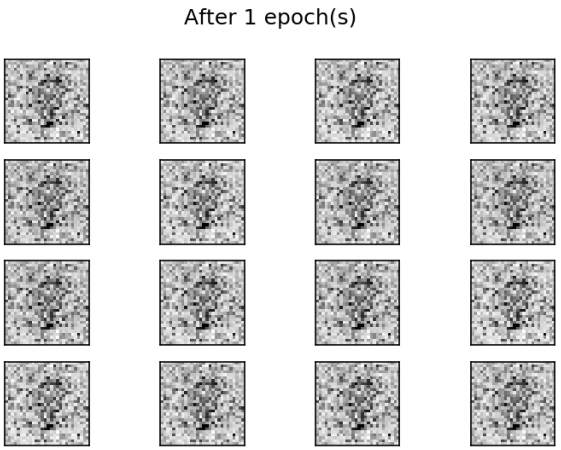
- **Generative adversarial network**
- **Diffusion Models**
- **Variational Autoencoder**



Generative Adversarial Network (GAN)



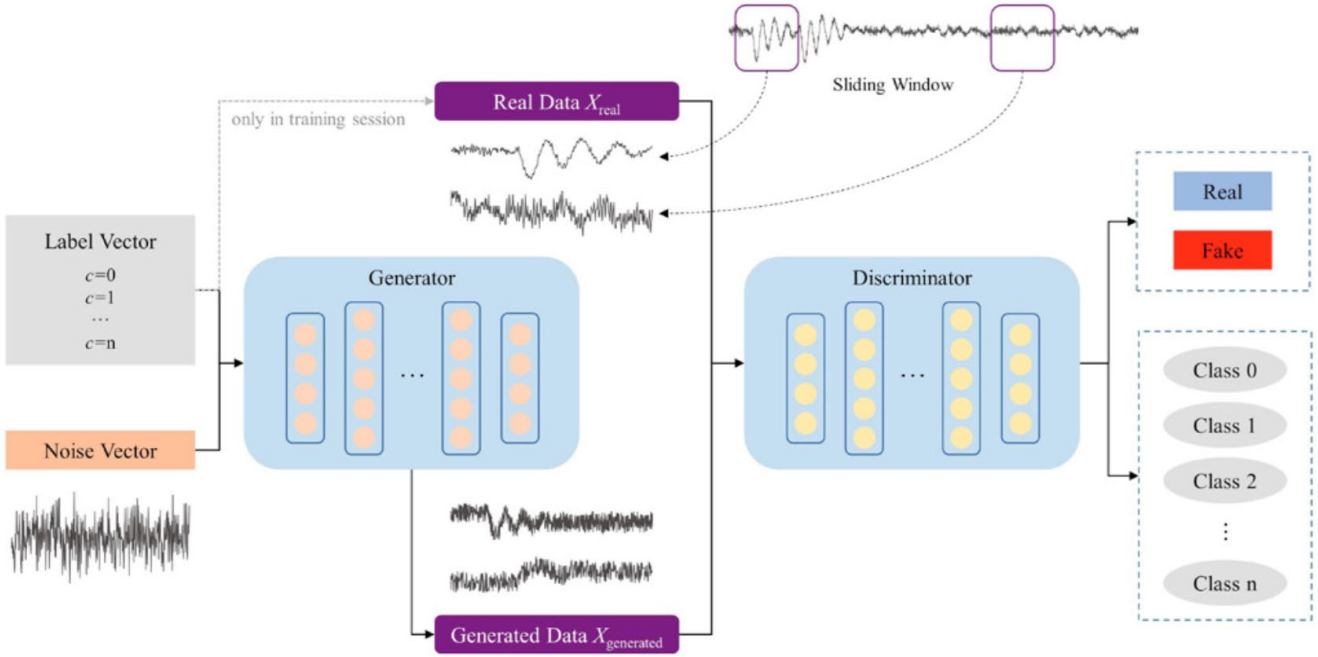
Overview of GAN's architecture
(credit: sthalles.github.io)



Visualize the evolution of the training
(credit: realpython.com)

- Invented by Ian Goodfellow and colleagues in 2014, GANs use two neural networks in opposition to generate highly realistic outputs, spurring major advances in generative AI for image creation.
- Inspired by game theory with two competing neural networks.
- A generator creates fake data while a discriminator tries to detect fakes, pushing the generator to produce increasingly realistic outputs.

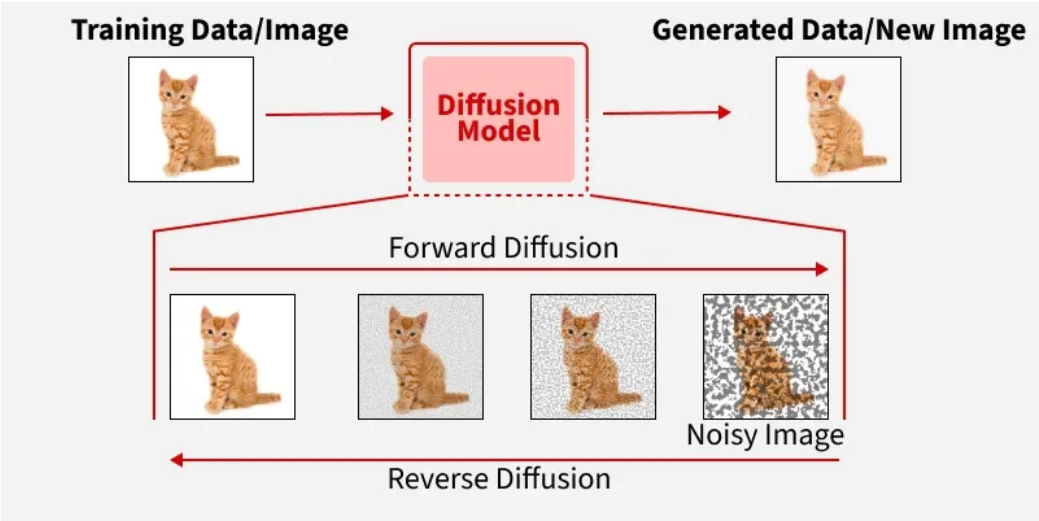
Application of GAN in Chemical Process Fault Diagnosis



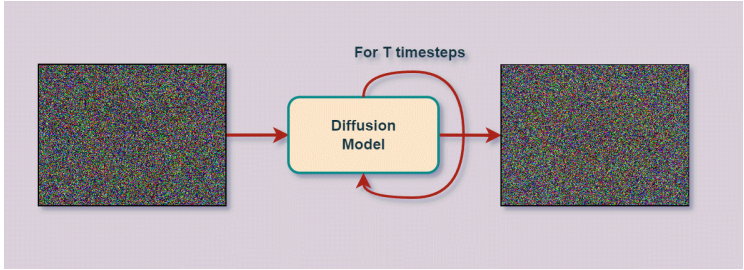
- In chemical plants, faults can be rare or expensive to reproduce, so we often don't have enough data to train traditional models. GANs can generate realistic synthetic fault data, helping balance the dataset and improve diagnosis accuracy even with few real samples.
- This model combines a generator that creates fake but realistic fault signals and a discriminator that learns to tell real from fake while also classifying faults.

Qin, Ruoshi, and Jinsong Zhao. "High-efficiency generative adversarial network model for chemical process fault diagnosis." *IFAC-PapersOnLine* 55, no. 7 (2022): 732-737.

Diffusion Model



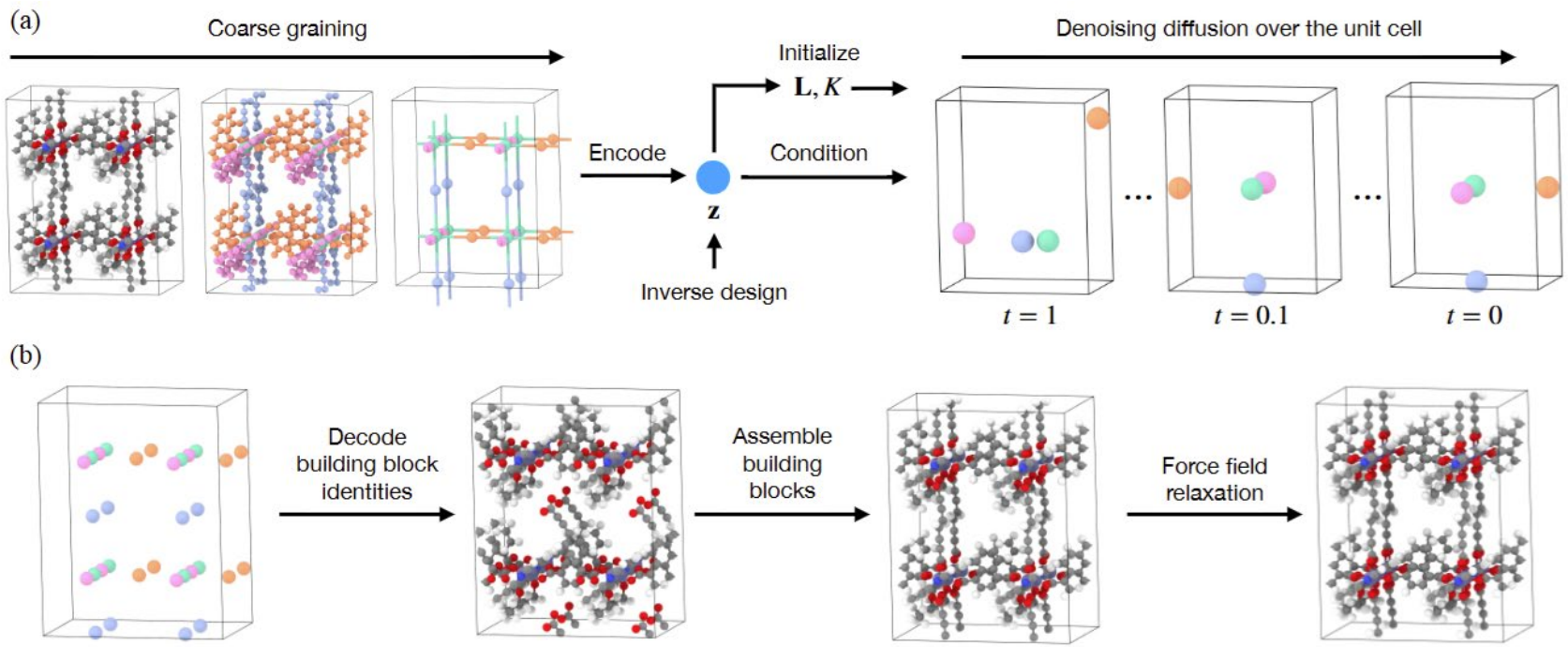
Diffusion model
(credit: geeksforgeeks.org)



Denoising process
(credit: learnopencv.com)

- Invented by Sohl-Dickstein and colleagues in 2015, diffusion models introduced iterative denoising procedures that have become foundational for controllable and high-quality image generation
- Inspired by non-equilibrium thermodynamics and stochastic diffusion.
- Progressively adds noise to data and then learns to reverse the process to generate new realistic samples.

Application of Diffusion Models in Material Discovery

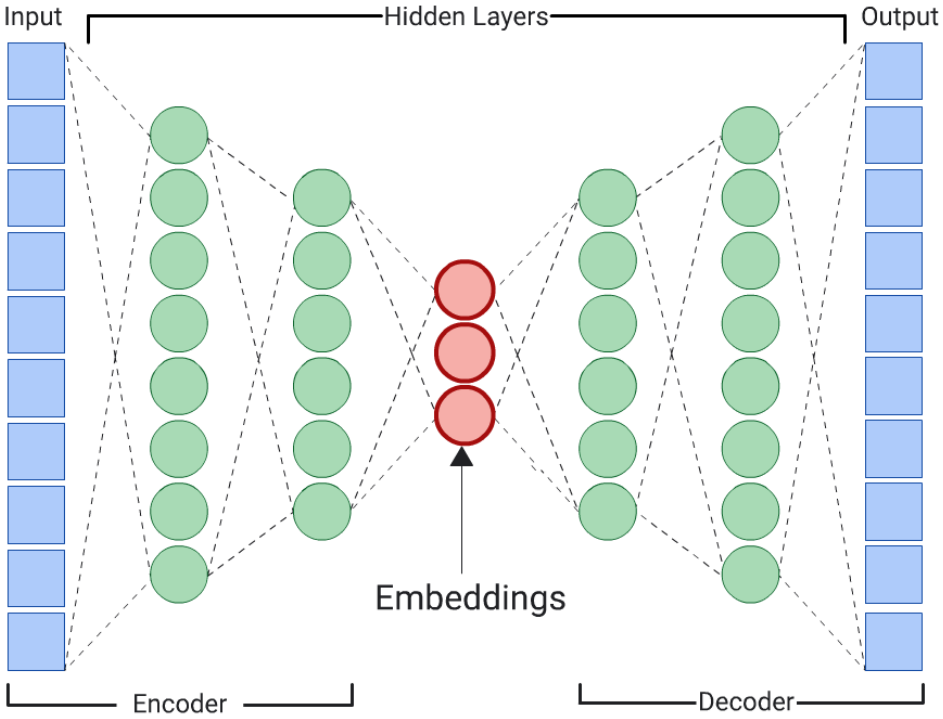


- MOFDiff encodes a coarse-grained (CG) representation of MOF structures and decodes CG MOF structures with a denoising diffusion process.
- The new all-atom MOF structure is recovered from denoised coarse-grained representation.

Fu, Xiang, Tian Xie, Andrew S. Rosen, Tommi Jaakkola, and Jake Smith. "Mofdiff: Coarse-grained diffusion for metal-organic framework design." arXiv preprint arXiv:2310.10732 (2023).

Autoencoder

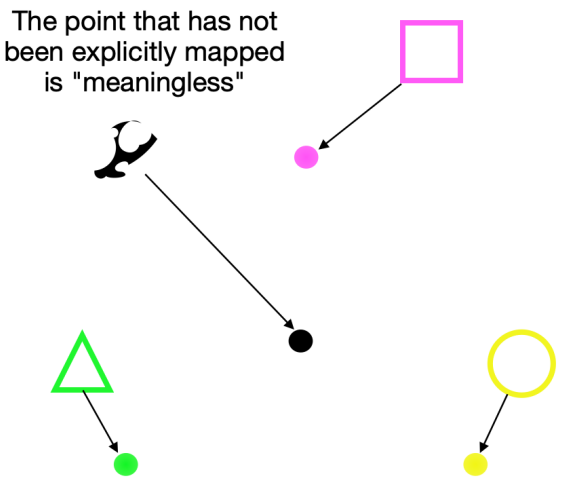
- First introduced by Rumelhart, Hinton, and Williams in 1985, autoencoders laid the foundation for unsupervised neural network architectures that compress and reconstruct data by learning meaningful low-dimensional representations.
- Encodes input data into a latent space and then decodes samples from it to generate new, realistic data.

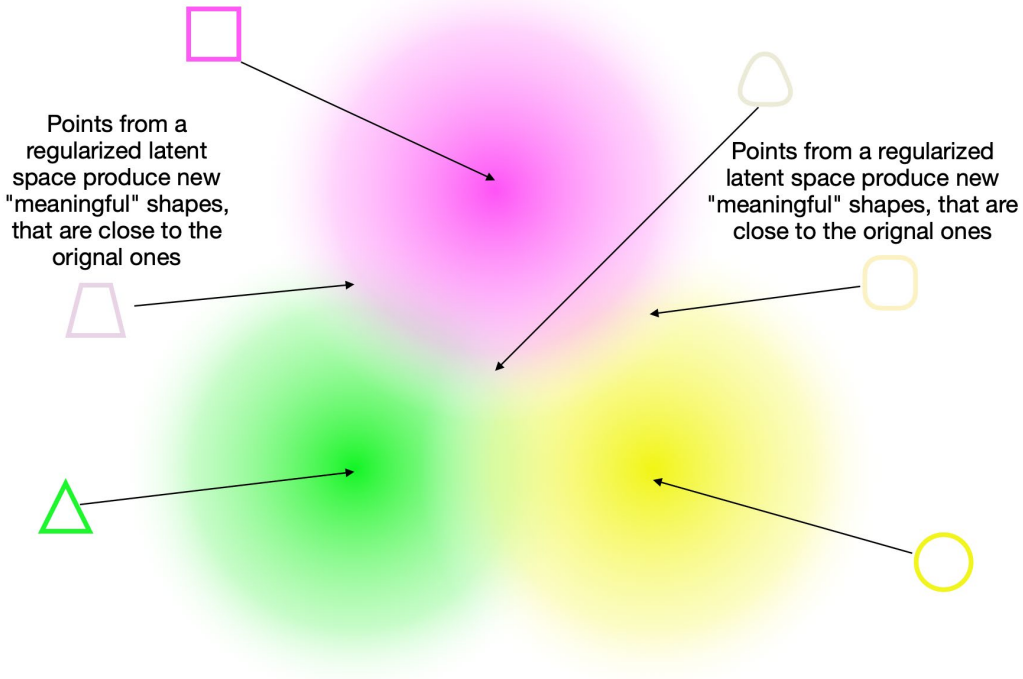


Schematics of autoencoder
(image credit: Google)

"Learning Internal Representations by Error Propagation", https://apps.dtic.mil/sti/tr/pdf/ADA164453.pdf

Discrete Latent Space is not Suitable for New Data Generation



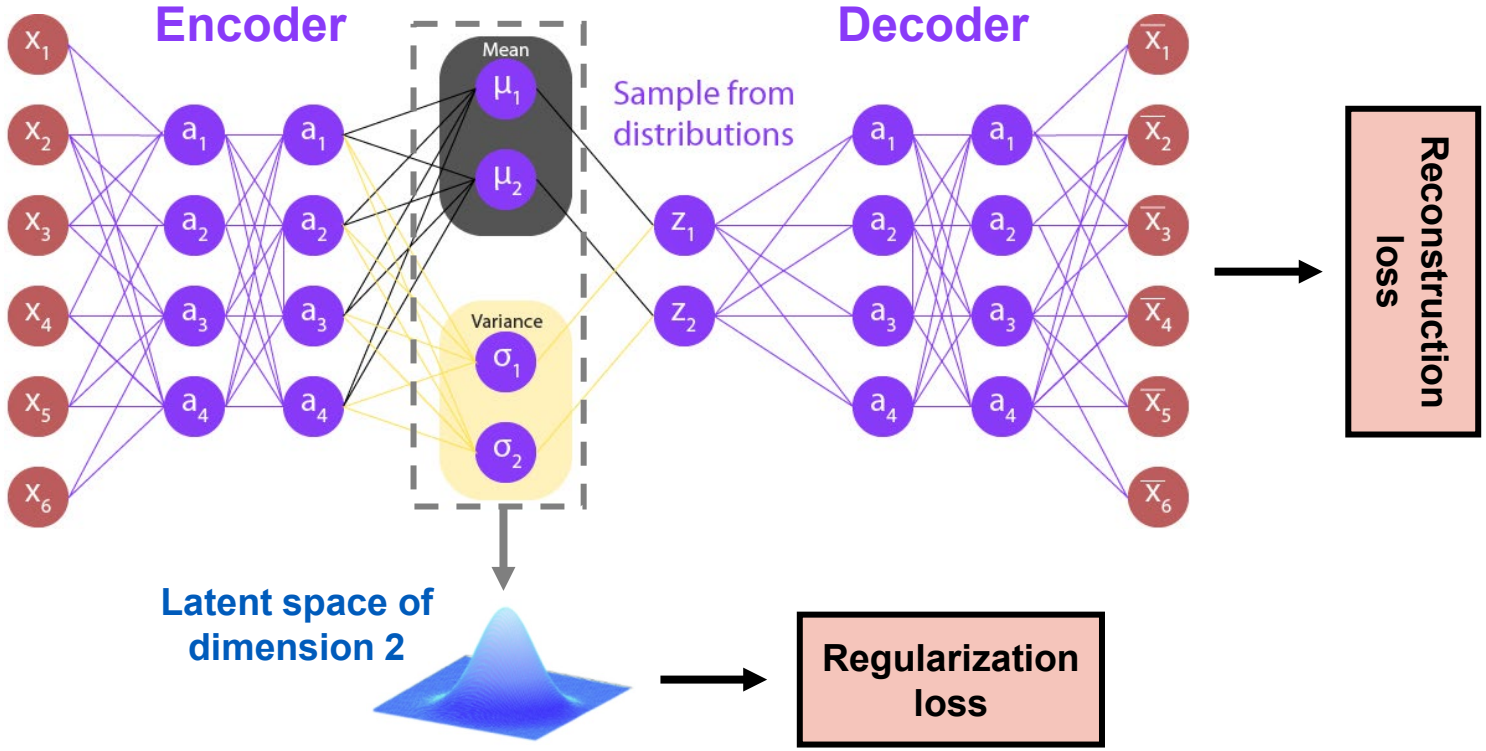


- Data mapped as **discrete** points in the latent space (e.g., typical autoencoder)
- Not suitable for generating new data.

- Data mapped as **distributions** (e.g., **variational autoencoder**)
- Latent space is continuous and regularized, suitable for new data generation.

Image credit: towardsdatascience.com

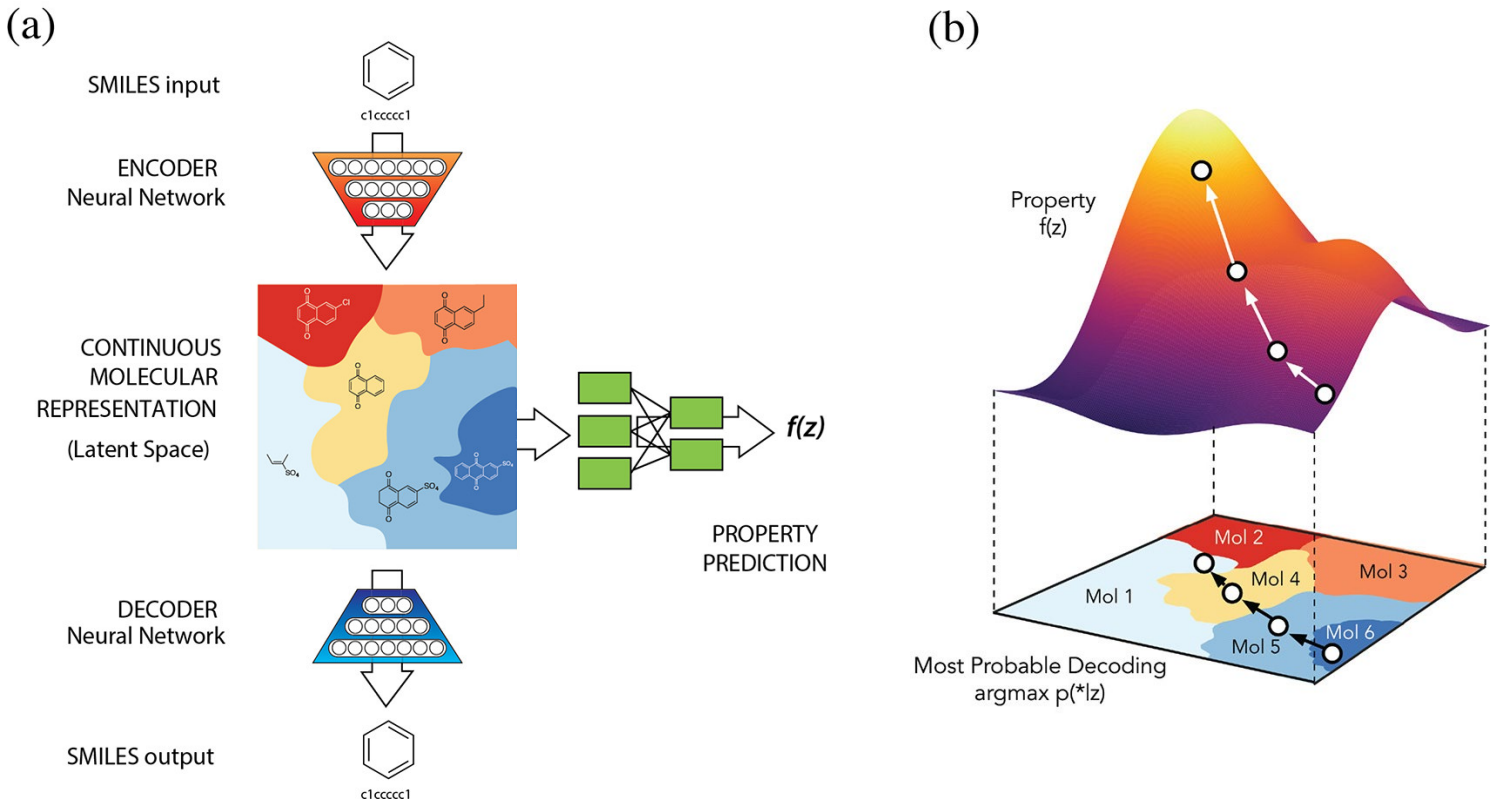
Variational Autoencoder (VAE)



- Kingma & Welling (2013) introduced VAEs, a probabilistic version of autoencoder for generation.
- Encoder maps input $\mathbf{x}$ to regularized, continuous latent "clouds" (i.e., Gaussian distribution) with mean $\boldsymbol{\mu}(\mathbf{x})$ and variance $\boldsymbol{\sigma}(\mathbf{x})$, with vector dimension corresponding to that of the latent space.
- Decoder samples latent vector randomly from learned distribution and reconstructs the input $\bar{\mathbf{x}}$.
- Loss minimizes the Reconstruction and regularization to smooth and organize the latent space.

Image adapted from Geeksforgeeks

VAE Applications: Molecular Discovery



a) A diagram of the autoencoder used for molecular design, including the joint property prediction model. Starting from a discrete molecular representation, such as a SMILES string, the encoder network converts each molecule into a vector in the latent space, which is effectively a continuous molecular representation. Given a point in the latent space, the decoder network produces a corresponding SMILES string. A mutilayer perceptron network estimates the value of target properties associated with each molecule.

b) Gradient-based optimization in continuous latent space. After training a surrogate model f(z) to predict the properties of molecules based on their latent representation z, we can optimize f(z) with respect to z to find new latent representations expected to have high values of desired properties.

Gómez-Bombarelli, Rafael, et al. "Automatic chemical design using a data-driven continuous representation of molecules." *ACS Central Science* 4.2 (2018): 268-276.

Hands-on: VAE for Drug Discovery

See Google Colab Notebook